

Real analysis general exam, January 2020

Problem 1

Let μ be the Lebesgue measure on $\mathbb{R}$. For a Lebesgue measurable set $A \subset [0, 1]$, is it true that

- (a) $\mu(A) = \sup_{U \subset A, U \text{ open}} \mu(U)$? If true, prove this. If false, give a counterexample.
(b) $\mu(A) = \inf_{U \supset A, U \text{ open}} \mu(U)$? If true, prove this. If false, give a counterexample.

Problem 2

Find a polynomial $P(x)$ of degree at most 3 such that $\int_{-1}^1 |x^4 - P(x)|^2 dx$ is minimal.

Problem 3

Let X be a compact metric space, and $C(X)$ be the space of all real-valued continuous functions on X with the supremum norm. Assume that the subset $\mathcal{A} \subset C(X)$ satisfies the following properties:

- (algebra) For all $f, g \in \mathcal{A}$ and $\alpha, \beta \in \mathbb{R}$ we have $\alpha f + \beta g \in \mathcal{A}$ and $fg \in \mathcal{A}$.
- (separates points) For any $x \neq y$ from X there exists a function $f \in \mathcal{A}$ such that $f(x) \neq f(y)$.

This question has two parts:

- (a) Show by example that $\mathcal{A}$ need not be dense in $C(X)$, explicitly checking all the properties of your example $\mathcal{A}$.
(b) In order to conclude that $\mathcal{A}$ is dense by Stone-Weierstrass Theorem, what additional condition(s) should be added?

Problem 4

Let μ be a measure on $(\mathbb{R}, \mathcal{B})$, where $\mathcal{B}$ is the Borel σ -algebra. Let $\mu(\mathbb{R}) = 1$. Next, let $\mathcal{F} \subset \mathcal{B}$ be the sub- σ -algebra of symmetric Borel sets, that is, $\mathcal{F}$ generated by all intervals of the form $(-a, a)$ with $a > 0$.

Let $f \in L^1(\mathbb{R}, \mathcal{B}, \mu)$. Find a function g such that:

- (a) $g \in L^1(\mathbb{R}, \mathcal{F}, \mu)$ (in particular, g is $\mathcal{F}$ -measurable).
(b) For all $E \in \mathcal{F}$ we have $\int_E g d\mu = \int_E f d\mu$.

Problem 5

Let μ be a finite measure on some measurable space $(X, \mathcal{F})$.

Show that a sequence of $\mathcal{F}$ -measurable functions f_n converges to a function f in measure if and only if $\int_X \min \{1, |f_n - f|\} \mu(dx) \rightarrow 0$ as $n \rightarrow +\infty$.